

TABLE I
UTILITY-SCALE ENERGY STORAGE APPLICATIONS

Application	Capacity requirement	Classification	Discharge cycles per year	Applicable technologies
Peak pricing arbitrage	4-6 hours	Bulk storage	200-400	Advanced batteries, compressed-air energy storage (CAES), pumped storage
Generation capacity	2-6 hours	Bulk storage	200-600	Advanced batteries, CAES, pumped storage
Transmission and distribution asset capacity	2-4 hours	Bulk storage	201-600	Advanced batteries, CAES
Frequency regulation	1-15 mins	Ancillary/power services	1000-20,000	Advanced batteries, flywheels, ultracapacitors
Volt/VAR support	1-15 mins	Ancillary/power services	1000-20,000	Li-ion, advanced lead-acid, flywheels, ultracapacitors
Renewables ramping/smoothing	1-15 mins	Ancillary/power services	500-10,000	Advanced batteries, flywheels, ultracapacitors

Source: Navigant Research

become one of the largest markets for advanced energy storage technologies but also a global hub for manufacturing various energy storage technologies due to an attractive domestic market and strengthening of business environment. With Make in India programme, the government of India aims to encourage various technology developers to set up their manufacturing base in India. Although India missed the opportunity to set up mainstream manufacturing for solar, energy storage is an upcoming industry where Indian manufacturers can play a key role in the global supply chain.

Market dynamics

India's energy storage potential is based on the combination of energy

TABLE II
BTM OR OFF-GRID ENERGY STORAGE APPLICATIONS

Market drivers	Customer applications	Description/benefit
Rising electricity rates, increasing electric vehicle (EV) use, increasing energy management system use	Demand charge reduction	Respond automatically to building load spikes
	Time-of-use (TOU) energy bill management	Manage charging and discharging based on retail electricity rates
Increasing solar PV installations	Onsite generation for self-consumption	Maximise consumption of onsite generation, primarily solar PV—reduced electricity expenses
Need for resiliency/power quality	Backup power/improved power quality	Protect sensitive equipment from power-quality fluctuations/outages
Grid stability concerns and capacity needs	Ancillary services	Provide frequency regulation, voltage support, electric supply reserve capacity, etc—improved efficiency of centralised generation, smoother integration of variable generation
	Demand response	Manage charging and discharging based on retail electricity rates
New utility infrastructure needs	Transmission and distribution investment deferral	Limit investments in new infrastructure through reduced peak demand

Source: Navigant Research

Energy storage technologies

Broadly, energy storage technologies can be grouped into electrochemical and non-electrochemical types. The most common energy storage technologies are:

1. Advanced lead-acid batteries
2. Flow batteries
3. Lithium-ion (Li-ion) batteries
4. Nickel-based batteries
5. Sodium-based batteries
6. Other emerging electrochemistries
7. Pumped hydroelectric
8. Compressed-air energy storage
9. Thermal storage
10. Gravity-based storage systems
11. Flywheel
12. Ultra-capacitors
13. Superconducting magnetic energy storage

(Source: IEA)

resources, historical physical infrastructure and electricity market structure, regulatory framework, population demographics, energy-demand patterns and trends, and general grid architecture and condition. The performance of these fundamental factors creates the demand for new products and services, and energy storage is increasingly being sought to meet these emerging requirements.

The Indian energy storage providers are gearing up with large-scale pilot projects. While lack of reliable grid power was a driver for traditional energy storage systems, ambitious renewable energy goals, growth in

electric vehicle sales, and need for reduction in diesel usage will boost the sales of energy storage systems throughout the country. The government of India has set up 175GW target for renewable energy deployment by 2022, and it is evident that energy storage technologies will play a key role in achieving this goal.

While specific drivers for energy storage markets vary with time, the overall market can be bucketed under three major segments:

1. Utility-scale or grid-tied applications refer to systems installed on transmission or distribution networks providing services to grid operators (Table I)